

University of Waterloo
Faculty of Engineering
Department of Electrical and Computer Engineering

Project Specifications and Risk Assessment

EchoMemory

Group 2026.50

Prepared by
Ekaum Singh

Prerak Jiwane
Anderson Yang
Eric Wang
James Chung
Raghav Verma

Consultant: Dr Otman Basir
Jun 8, 2027

Table of Contents

1. High-Level Project Description.....	3
1.1 Motivation.....	3
1.2 Project Objective.....	3
1.3 Block Diagram.....	5
2. Project Specifications.....	6
2.1 Functional Specifications.....	6
2.2 Non-functional Specifications.....	8
3. Risk Assessment.....	9
4. References.....	11

1. High-Level Project Description

1.1 Motivation

As of 2025, approximately 771,939 Canadians are living with dementia, with Alzheimer's disease accounting for 60–80% of cases [1]. A defining challenge for these patients is the loss of ability to independently manage daily activities at home including following routines, staying focused on tasks, and adhering to medication schedules. Research shows that even mild cognitive impairment increases the risk of medication non-adherence by more than four times compared to cognitively intact individuals [2], contributing to worsening symptoms and preventable hospitalizations. An ideal assistive system would monitor the patient's daily activities in real time, provide gentle reminders without requiring active input from the user, and reduce the supervision burden on caregivers.

1.2 Project Objective

The objective of this project is to design a wearable assistive system to assist Alzheimer's patients living at home in independently managing daily living activities — such as medication adherence and routine-following — by providing real-time, context-aware reminders and commentary without requiring active input from the user or continuous internet connectivity.

1.3 Block Diagram

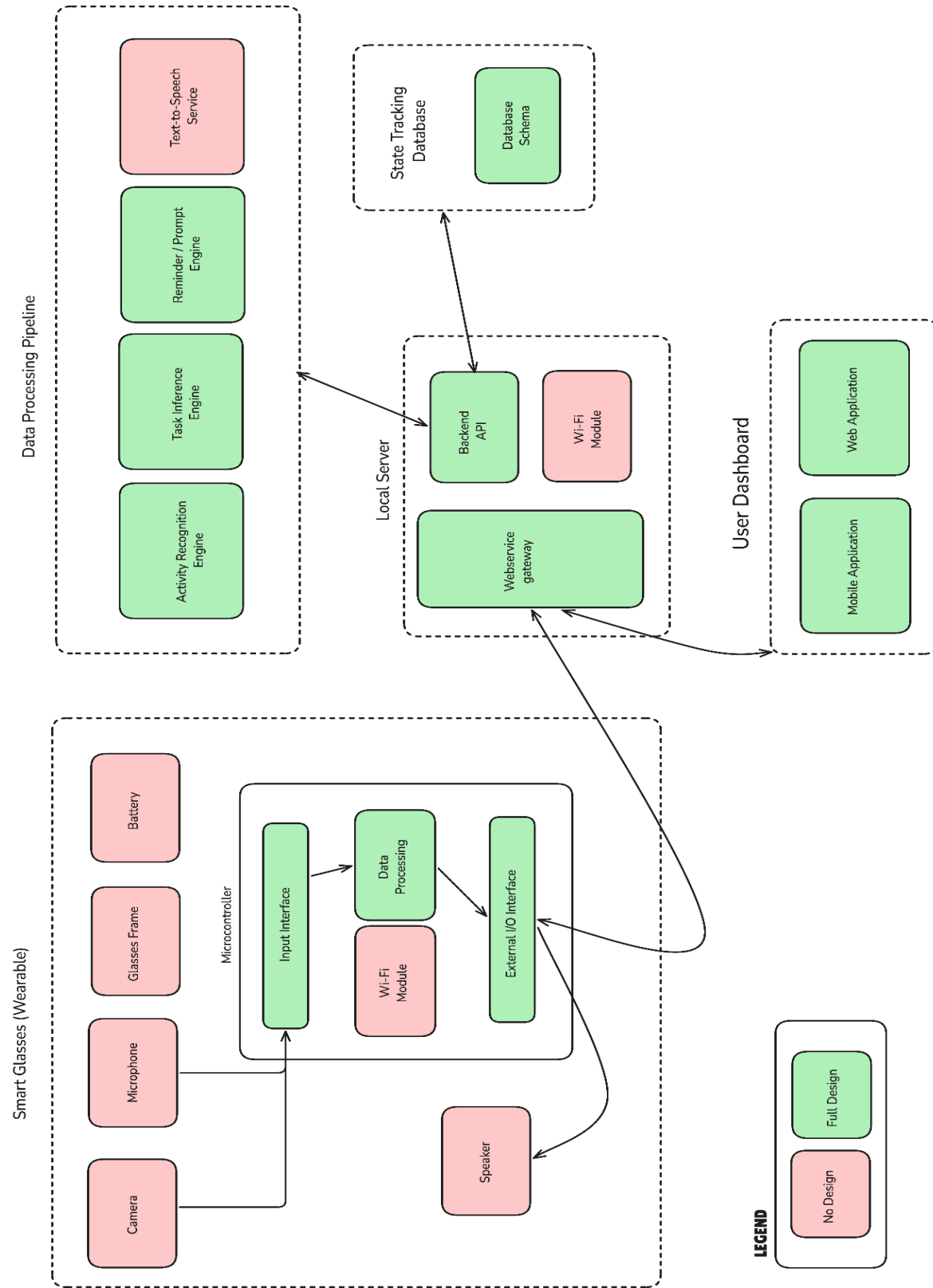


Figure 1-1. Block diagram overview of the various subcomponents

2. Project Specifications

The project specifications are classified as either functional or non-functional specifications. The functional specifications describe what the system does while the non-functional specifications describe characteristics of the system. Each specification is further sub-categorized as essential or non-essential. Essential specifications are those that must be met for the design to be satisfactory. Non-essential specifications extend the final product beyond the desired scope when they are met. They are non-critical aspects of the system, and the system will still function properly if they are not met. Each specification is rated as essential or non-essential based on importance to the overall design.

2.1 Functional Specifications:

Essential

Specification	Subsystem	Description
Video Capture and Streaming	Smart Glasses Wearable	The camera must capture video at a minimum resolution of 720p and 15 fps under indoor lighting conditions.
Task state tracking	Data Processing Pipeline	Segment the user's actions into a sequence of expected sub-activities and register a completed task once a defined sequence of sub-activities is detected, with a minimum accuracy of 80% under controlled conditions.
Wireless Data Transfer	Microcontroller	Stream camera data from the glasses to the host machine and receive the returned prediction with a round-trip transfer time under 500 ms, sufficient to meet the overall audio-feedback latency target.
Operational Runtime & Recharge	Smart Glasses Frame	Power the microcontroller from an onboard rechargeable

		battery. Recharge from empty to full in under 2 hours.
Audio feedback	Smart Glasses Frame	Deliver an audio reminder appropriate to the user's current activity within the system latency budget after the triggering condition is detected.

Non-Essential

Specification	Subsystem	Description
Data Visualization	User Dashboard	Present a daily summary of detected activities to the user and caregiver to support care decisions.
User Confirmation	Smart Glasses Frame	The user can confirm whether or not the model's predicted activity is accurate, providing feedback to finetune future predictions.

2.2 Non-functional Specifications:

Essential

Specification	Subsystem	Description
Local processing privacy	Local server	Ensures all captured video frames and streaming media packets are processed and decoded within the local area network (LAN). No raw image data is permitted to transit outside to external wide area network (WAN) IP

		addresses.
Database query latency	State tracking database	To prevent an outdated backlog in the data processing pipeline for detected tasks, the database must optimize read/write queries so that fetching or updating a user's activity state history takes less than 100 ms.
Model inference throughput	Data processing pipeline	The Activity Recognition Engine running on the edge server must process the incoming low-frame-rate video feed at a minimum throughput of 15 frames per second (FPS) to ensure real-time visual tracking of physical human movements.
Low power consumption	Microcontroller	Optimizes the firmware execution and sleep states of the microcontroller to ensure the wearable glasses achieve a minimum baseline operational runtime of 6-7 hours of continuous streaming on a single charge.
Low audio latency	Data processing pipeline	Restricts the total software loop delay from the initial camera frame capture on the glasses, through the edge server computer vision inference to the delivery of the reminder back to the glasses speaker, to under 1.5 seconds.
Wearing safety	Smart glasses frame	Restricts component power draw so that the outer casing temperatures touching the

		user's face never exceed 40 celsius under peak processing loads, eliminating any risk of skin irritation or thermal burns.
--	--	--

Non-Essential

Specification	Subsystem	Description
Lightweight	Smart glasses frame	Limits the total added physical mass of the hardware components (microcontroller, camera module, speaker, and battery) to under 60 grams to minimize nose and ear fatigue during prolonged daily use.
Microcontroller dimensions	Smart glasses frame	Caps the physical footprint of the microcontroller board and its custom 3D-printed enclosure so it sits flush against the temple arm without obstructing the user's field of view.
Dashboard page loading	User dashboard	The Web / Mobile Application dashboard interfaces should achieve a localized page load and UI rendering refresh speed of under 2 seconds upon caregiver access.

3. Risk Assessment

Table 4 below summarizes the most significant technical, logistical, and operational risks that could prevent the successful completion of the smart glasses project, along with strategic mitigation plans for each.

Table 4. Risk Assessment

Risk: Edge-to-Glasses Network Latency and Jitter
Impact: High
Probability: High
Situation: The project relies on streaming real-time video data from the glasses to a local edge server and receiving immediate audio feedback. If local Wi-Fi or network latency spikes, or if packet drop occurs, the system will fail its non-functional requirement for low-latency audio feedback. This renders the real-time activity tracking and prompt interventions useless for an Alzheimer's patient.
Remediation: The team will prioritize optimizing the video compression pipeline to drastically minimize bandwidth requirements. Additionally, a fallback mechanism will be implemented on the glasses' firmware to handle network drops gracefully, queueing critical state requests or playing a local "connection lost" audio alert instead of hanging indefinitely.

Risk: Activity State Tracking Edge Cases (Context Drift)
Impact: High
Probability: Medium
Situation: Alzheimer's patients may exhibit erratic behavior or skip critical steps in a routine (e.g., picking up medication but putting it right back down without swallowing it). The tracking algorithm might misinterpret these segmented, non-linear activities, leading to incorrect context building, missed reminders, or confusing, repetitive audio tracking prompts.
Remediation: The state tracking system will be designed defensively using probabilistic state models rather than rigid, linear step-checking logic. The team will prioritize validating a single, high-confidence task first (like medication tracking) and establish a safe "reset/unknown state" if user actions become too ambiguous to interpret.

Risk: Physical Bottleneck of a Single Prototype with a Large Team
Impact: High
Probability: High
Situation: With a large group of 6 members working on a highly integrated project, having only one physical prototype of the smart glasses creates a major development bottleneck. Hardware components, firmware calibration, and physical casing adjustments cannot be worked on simultaneously by multiple people. As testing deadlines approach, a lack of access to the physical device will completely stall individual progress on features like activity recognition and audio feedback.
Remediation: We will establish an automated CI/CD pipeline on the edge server. This allows team members to push software updates remotely to the server, where they can be tested against pre-recorded user video logs without needing the physical glasses in hand. OR A strict, shared calendar schedule will be implemented, mandating 48-hour rotations for who has possession of the physical prototype.

4. References

[1]Alzheimer Society of Canada. (2025). Dementia numbers in Canada.
<https://alzheimer.ca/en/about-dementia/what-dementia/dementia-numbers-canada>

[2]Black, A. et al. (2010). Medication adherence in healthy elders: small cognitive changes make a big difference.<https://pmc.ncbi.nlm.nih.gov/articles/PMC2910516/>